26T1 Week10

Course Content Revision!

✸ Software Design & Architecture ✸

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agenda

- Serverless Architecture
- Kahoot Quiz
- Revision Questions (Code/Design Smells, Design Patterns, Architectures, etc.)

- Congrats on finishing all your assignments for the term!
- Exam environment sample exam this week
- myExperience pretty please

some admin

serverless architecture

- AKA Function-as-a-Service (FaaS)
- Servers do exist, "serverless" means you don't manage them
  - Enables **building and running applications without managing infrastructure**
- Developers just deploy individual functions → cloud provider handles the rest (i.e. manages server allocation dynamically)
- Cloud providers bill for execution time ('micro-billing') to the millisecond
  - Users are charged for the actual resources consumed during code execution rather than for pre-provisioned, idle server capacity
- The small, stateless functions sleep until triggered by events (e.g. file upload, HTTP request)

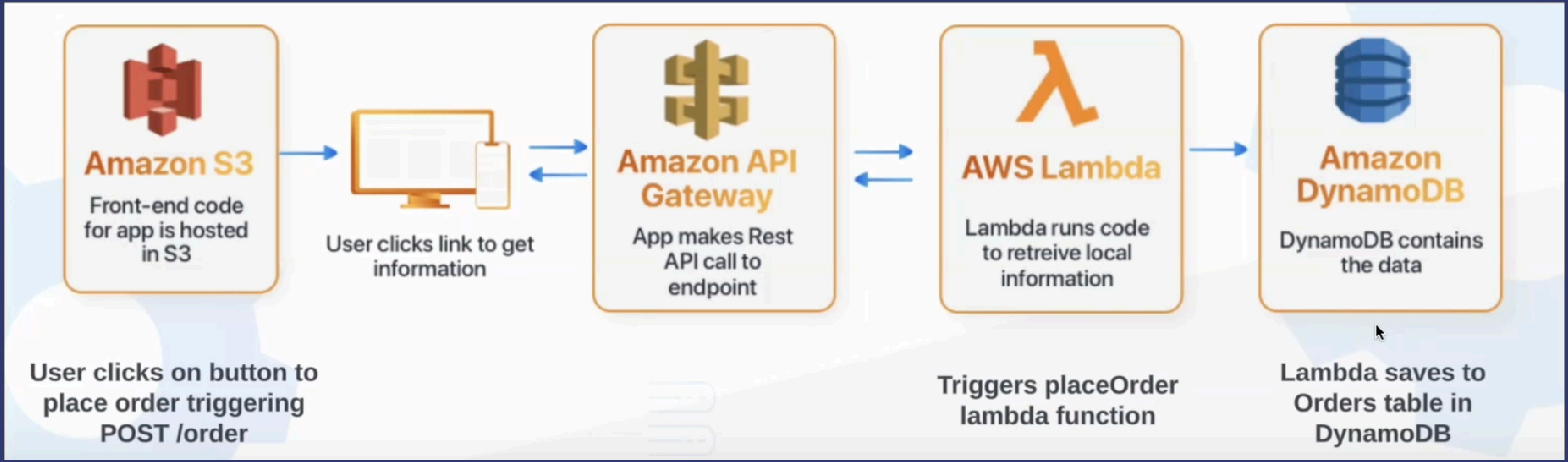
Trigger → API Gateway → Function Executes → Interacts with Services → Returns Result

serverless architecture

<u>When use it?</u>

Sporadic, event-triggered, stateless tasks, not always-on workloads

- Form submission → store to DB
- Image upload → auto-resize
- Firebase DB change → push notification to users
- E-commerce → on-demand inventory updates



serverless architecture

Strengths

- **Auto-scaling**: handles thousands of concurrent requests instantly, no manual config; resources adjusted based on real-time demand
- **Cost optimisation**: pay only for execution time, not idle time
- **High availability**: distributed across multiple availability zones automatically
- **Short-lived**: designed for tasks that complete fast
- **Faster dev cycle**: teams focus on business logic, not server maintenance. Independent function updates with redeploying the entire app

Weaknesses

- **Cold starts**: Takes time to initiate internal resources for first function request
- **Limited control:** Reliant on the provider and their ecosystem
- **Debugging complexity**: Functions run in *short-lived instances without persistent state* (statelessness allows for reliable, independent scaling), making it difficult to replicate problems and harder to track request flows across functions

serverless example

Interactive Diagram: Ilograph

revision!

Exam Prep

Kahoot Revision Quiz!

A quick quiz on Java, OOP, design patterns and more.

✦ finding patterns

In groups, determine a possible pattern that could be used to solve each of the problems.

1. Sorting collections of records in different orders.
2. Listing the contents of a file system.
3. Updating a UI component when the state of a program changes.
4. A frozen yogurt shop model which alters the cost and weight of a bowl of frozen yogurt based on the toppings that customers choose to add before checkout.
5. A game where enemy guards are alerted when the player makes a loud noise.
6. A cross-platform application that generates different types of UI components (buttons, menus, dialogs) to match the look and feel of each operating system (Windows, macOS, Linux).
7. A text editor application where you can apply multiple formatting options (bold, italic, underline, highlight, different fonts) to text, with the ability to combine these effects and add or remove them dynamically

code & design smells

Discuss the following examples. Identify the code smells and any underlying design problems associated with them.

Mark, Bill and Jeff are working on a **PetShop application**. The PetShop has functionality to feed, clean and exercise different types of animals.

Mark notices that **each time he adds** a new species of animal to his system, he also has to **rewrite all the methods** in the PetShop so it can take care of the new animal.

identify code/design smell(s)

```java
public class Person {
  private String firstName;
  private String lastName;
  private int age;
  private int birthDay;
  private int birthMonth;
  private int birthYear;
  private String streetAddress;
  private String suburb;
  private String city;
  private String country;
  private int postcode;

  public Person(String firstName, String lastName, int age, int birthDay, int birthMonth, int birthYear,
      String streetAddress, String suburb, String city, String country, int postcode) {
    this.firstName = firstName;
    this.lastName = lastName;
    this.age = age;
    this.birthDay = birthDay;
    this.birthMonth = birthMonth;
    this.birthYear = birthYear;
    this.streetAddress = streetAddress;
    this.suburb = suburb;
    this.city = city;
    this.country = country;
    this.postcode = postcode;
  }
  // Some various methods below
  // ....
}
```

✴ identify the code/design smell(s)

```java
public class MathLibrary {
  List<Book> books;

  int sumTitles() {
    int total = 0
    for (Book b : books) {
      total += b.title.titleLength;
    }
    return total;
  }
}

public class Book {
  Title title; // Our system just models books as titles (content doesn't matter)
}

public class Title {
  int titleLength;

  int getTitleLength() {
    return titleLength;
  }

  void setTitleLength(int tL) {
    titleLength = tL;
  }
}
```

identify the code or design smell(s)

architecture revision

For each scenario, determine which style of software architecture would best suit the requirements.

serverless questions

✦ ✦ ✦

Describe a scenario where a serverless function would be used to handle user-submitted form data.

1. User submits a form on a website (e.g. a contact or feedback form)
2. Action triggers a serverless function via an API Gateway.
3. Function validates the data and stores it in a database.

- Serverless is ideal here because we only want it to run when the form is submitted.
- We don't know how consistent traffic will be (nor how much traffic we'll get). This reduces the cost and removes the need to manage a backend server for infrequent submissions.

serverless for form data

serverless questions

How could a serverless architecture be used to support an image upload feature on a website?

1. User uploads an image to a cloud storage service
2. Upload event can trigger a serverless function to automatically process the image. This could involve resizing or compressing the image.
3. Store the modified version in a new folder.

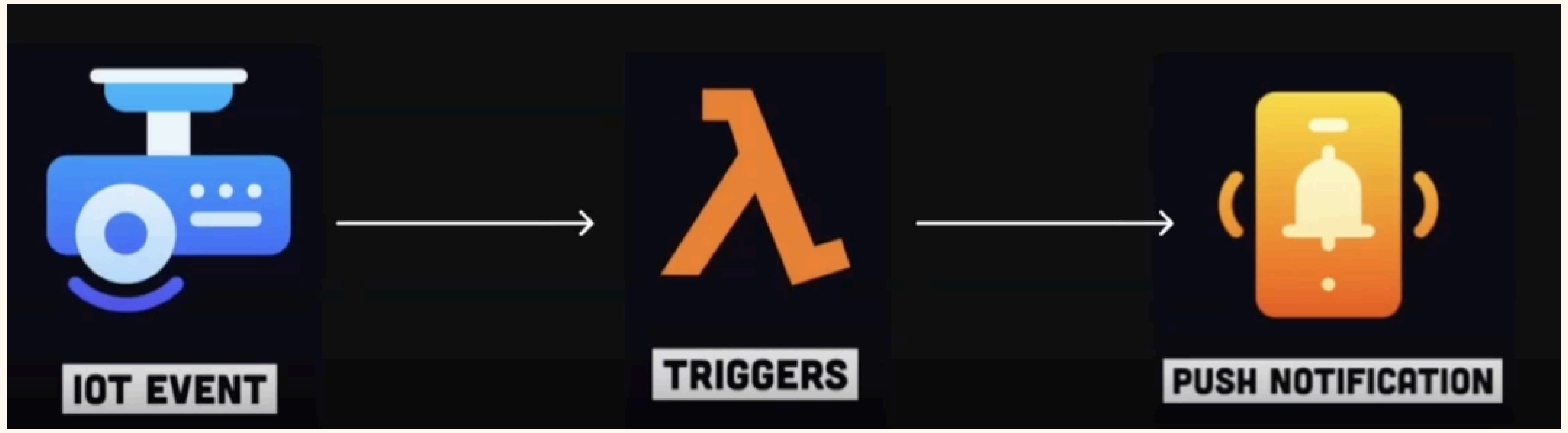
- Scales automatically, handles concurrent uploads efficiently, and avoids idle compute costs when no uploads occur (which we'd see if we used a server).

serverless for image upload

serverless questions

Explain how serverless functions could be used in an IoT health monitoring system.

serverless for IoT health monitoring ✦



- IoT devices (e.g. smart watches) can send user health data to the cloud periodically
    - This is why it might take a while to sync your fitness data between devices!
- Each incoming data packet can trigger a serverless function that checks the data for anomalies (e.g. dangerously high heart rate) and sends a push notification to the user.

serverless questions

✦ ✦ ✦

Why is a serverless approach useful for sending notifications in real-time chat or messaging applications?

serverless for real-time chats ✦

- Serverless functions can be triggered by changes in a database (e.g. a new message added) to instantly send push notifications to users.
- Since messages arrive unpredictably and traffic may spike, serverless ensures fast event-driven responses without needing dedicated servers running 24/7.

serverless questions

In what way can serverless functions improve scalability for an e-commerce site during high traffic periods like Black Friday?

serverless for real-time chats ✦

- Serverless functions automatically scale to handle a large number of concurrent requests
    - e.g each time a user places an order or views a product, a function can be triggered to update inventory or process payments.
- During high traffic events like Black Friday, this elasticity ensures smooth performance without provisioning infrastructure ahead of time, and you only pay for actual usage.

lab time!

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